

INC-2026-0903 - Duplicate Refund Application After Consumer Restart

Incident Report | Engineering

Document ID	INC-2026-0903	Seller ID	S003
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-09-06
Related Scenario ID	SC-007	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0903
Severity	SEV-2
Service	Payments - Refund Webhook Consumer
Component	Refund event processing
Occurred	2026-09-02 21:15 to 2026-09-03 02:40 IST
Detected	2026-09-03 17:20 IST
Mitigated	2026-09-04 16:30 IST
Resolved	2026-09-10 09:00 IST
Incident owner	Vikram Singh, Engineering
Seller affected	S003 - GreenCart Organics
Related ticket	TKT-2026-0903-336

A deployment restarted the refund webhook consumer. The broker redelivered events that had been processed but not acknowledged. The consumer applied refunds without an idempotency check, so 640 refunds were applied twice, refunding customers a total of INR 1,18,000 in excess.

2. Impact

Dimension	Detail
Orders refunded twice	640
Refund events processed	1,280 for 640 unique refunds
Excess refunded	INR 1,18,000
Time window	5 hours 25 minutes
Sellers affected	S003 predominantly; 3 orders across other accounts
Duplicate transactions reversed	412
Amounts written off	228 cases
Customer harm	None; customers received more, not less

The concentration on a single seller was structural rather than account-specific. GreenCart operates under a refund-on-report category exception for perishable goods: a customer reporting a quality problem is refunded without a physical return. This account therefore generates several times more refund events per order than any other seller, so a defect in refund processing surfaces here first and hardest.

The three affected orders on other accounts confirm the defect was platform-wide.

3. Detection

Detected by the seller, approximately 20 hours after the first duplicate.

GreenCart's payment reconciliation showed refund totals for 2 and 3 September at roughly double the expected value. The co-founder manually checked twelve orders, found two refund transactions against each, and raised the ticket at 17:20 IST on 3 September.

There was no platform-side control that would have detected this. Refund volume per account was not monitored against expected volume, and duplicate refund application produced no error: each individual refund succeeded.

4. Timeline

Time (IST)	Event
02 Sep 21:12	Deployment restarts the refund webhook consumer
02 Sep 21:15	Broker redelivers unacknowledged events; first duplicates applied
03 Sep 02:40	Last duplicate applied as redelivery backlog clears
03 Sep 17:20	Seller raises ticket after reconciliation
03 Sep 17:48	Escalated to Engineering and Operations
03 Sep 18:30	Duplicate refund transactions confirmed, 640 orders
03 Sep 20:10	Event redelivery confirmed as the mechanism
04 Sep 11:00	Seller call; recovery approach agreed by Operations
04 Sep 16:30	Idempotency store deployed; further duplication prevented
05 Sep 14:00	412 duplicate transactions reversed at the processor
08 Sep 10:00	228 residual cases written off
10 Sep 09:00	Refund duplication alert configured; incident closed

5. Symptoms

- Two refund transactions against a single order, with distinct transaction identifiers and identical amounts.
- Both transactions referencing the same refund event identifier.
- Timestamps clustered in a window beginning three minutes after a consumer restart.
- No errors in the refund consumer logs; every refund reported success.
- Refund volume for the account approximately double the expected count for the period.

The shared refund event identifier across the two transactions was the decisive evidence. It established that a single upstream event had been processed twice, rather than two refunds having been legitimately raised.

6. Affected Services

Service	Role	State
Refund Webhook Consumer	Applied refund events	Defective: no idempotency check
Event broker	Delivered refund events	Functioning as designed: at-least-once
Payment processor	Executed refund transactions	Functioning as designed
Settlement service	Recorded refunds against seller settlement	Correct, based on incorrect input
Deployment pipeline	Restarted the consumer	Functioning as designed

No component malfunctioned. Every component did exactly what it was built to do. The defect was an incorrect assumption about delivery semantics.

7. Investigation

7.1 The Mechanism

The event broker provides at-least-once delivery. A consumer receives an event, processes it, and acknowledges. If the consumer terminates after processing but before acknowledging, the broker redelivers on restart.

The deployment at 21:12 on 2 September performed a rolling restart. Consumer instances were terminated with a 30-second grace period. Events that had been processed within that window but not yet acknowledged were redelivered when the replacement instances started.

The consumer's refund application logic was:

1. Receive refund event.
2. Look up the order and the original payment.
3. Issue a refund to the payment processor.
4. Record the refund locally.
5. Acknowledge the event.

There is no step that asks whether this refund event has already been applied. A redelivered event therefore produced a second, entirely valid, refund transaction.

7.2 Why 640 And Not More

The redelivery window was bounded by the number of events in flight at restart. 640 events had been processed and not acknowledged across the consumer group at the moment of the rolling restart, which is consistent with the account's refund volume and the 30-second grace period.

The five-hour tail between 21:15 and 02:40 reflects the redelivery backlog being worked through alongside ongoing live traffic, not a five-hour window of new duplication.

7.3 Why This Was Not Caught Earlier

The consumer was written in 2023, before at-least-once handling was an explicit written requirement. The requirement now exists in two places: as ORD-3 in the S002 integration specification and as a design decision in the S001 order webhook receiver, which deduplicates on order identifier.

Both of those were written for order events. Nobody revisited the refund consumer against the same requirement. It had also never been restarted under load in a way that produced a meaningful redelivery volume, so the defect had no opportunity to surface.

8. Root Cause

The refund webhook consumer applied refund events without checking whether an event had already been processed. Because the event broker provides at-least-once delivery, a rolling restart during a deployment caused 640 processed-but-unacknowledged events to be redelivered and applied a second time, issuing duplicate refunds totalling INR 1,18,000 in excess.

9. Contributing Factors

- 1. **No idempotency check** on a financial operation consuming an at-least-once stream.
- 2. **Requirement not applied consistently.** At-least-once handling was required for order events and never audited across other consumers.
- 3. **No refund volume monitoring** per account against expected volume.
- 4. **Grace period shorter than the acknowledgement path**, making redelivery likely on any rolling restart.
- 5. **Deployment performed during the account's peak refund window**, evening being when quality reports are raised.
- 6. **Duplicate refunds are silent.** Each one succeeds; nothing fails.

10. Mitigation

The idempotency store was deployed at 16:30 on 4 September, which stopped any further duplication. Before that, no mitigation was applied: the duplication had already ended at 02:40 when the redelivery backlog cleared, so there was no ongoing harm to stop.

The substantive mitigation question was financial rather than technical, and it belonged to Operations: how to recover INR 1,18,000 from a cash- constrained seller who had not caused the error.

The decision was to recover across four settlement cycles beginning 15 September, with no interest or fee. Customers who received a duplicate refund were not asked to return it. Both the idempotency requirement and the staged recovery default are set by the Payment and Settlement Policy (POL-FIN-002) and executed under the Payment Recovery Runbook (OPS-RB-004).

11. Resolution

Action	Completed
Idempotency store keyed on refund event identifier, 30-day retention	2026-09-04 16:30
412 duplicate transactions reversed at the payment processor	2026-09-05 14:00
228 unsettled residual cases written off on proportionality grounds	2026-09-08 10:00
Consumer grace period extended beyond the acknowledgement path	2026-09-07

Action	Completed
Refund duplication alert on volume per account	2026-09-10 09:00
Audit of all consumers against at-least-once requirements	2026-09-19

The audit on 19 September examined every consumer of an at-least-once stream that performs a non-idempotent operation. It found two further consumers without idempotency protection, neither financial. Both were corrected.

12. Validation

Check	Method	Result
Duplicate application prevented	Same refund event replayed 50 times in staging	One refund applied; 49 rejected as duplicates
Rolling restart safe	Consumer restarted under synthetic load	No duplicates across 12,000 events
Retention window adequate	Redelivery simulated at 29 days	Correctly rejected as duplicate
Alert effective	Refund volume artificially doubled in staging	Alert raised within 4 minutes
No legitimate refunds blocked	Production observation, 14 days	Zero false rejections

The last check mattered. An idempotency store that is too aggressive would block legitimate second refunds on the same order, which do occur when a partial refund is followed by a further partial refund. Keying on event identifier rather than on order identifier avoids this.

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Idempotency store for refund event processing	Vikram Singh	2026-09-04	Complete
PA-2	Extend consumer grace period beyond acknowledgement	Vikram Singh	2026-09-07	Complete
PA-3	Refund duplication alerting per account	Vikram Singh	2026-09-10	Complete
PA-4	Audit all consumers for idempotency on non-idempotent operations	Vikram Singh	2026-09-19	Complete
PA-5	Idempotency required for any financial event consumer, in design review	Engineering	2026-09-30	Complete

ID	Action	Owner	Target	Status
PA-6	Avoid deployments during account-specific peak refund windows	Neha Gupta	2026-10-02	In progress

14. Follow-Up

The finding worth carrying forward is that at-least-once delivery had been treated as a property to handle where someone remembered, rather than as a platform-wide characteristic requiring a systematic control. PA-4 and PA-5 convert it into a design review gate.

Post-incident review held 6 September 2026. The review recorded that the seller's structural exposure to refund defects, arising from the refund-on-report category exception, should be noted in the account record so that future changes to refund processing consider this account first. That note was added to the S003 technical integration notes.